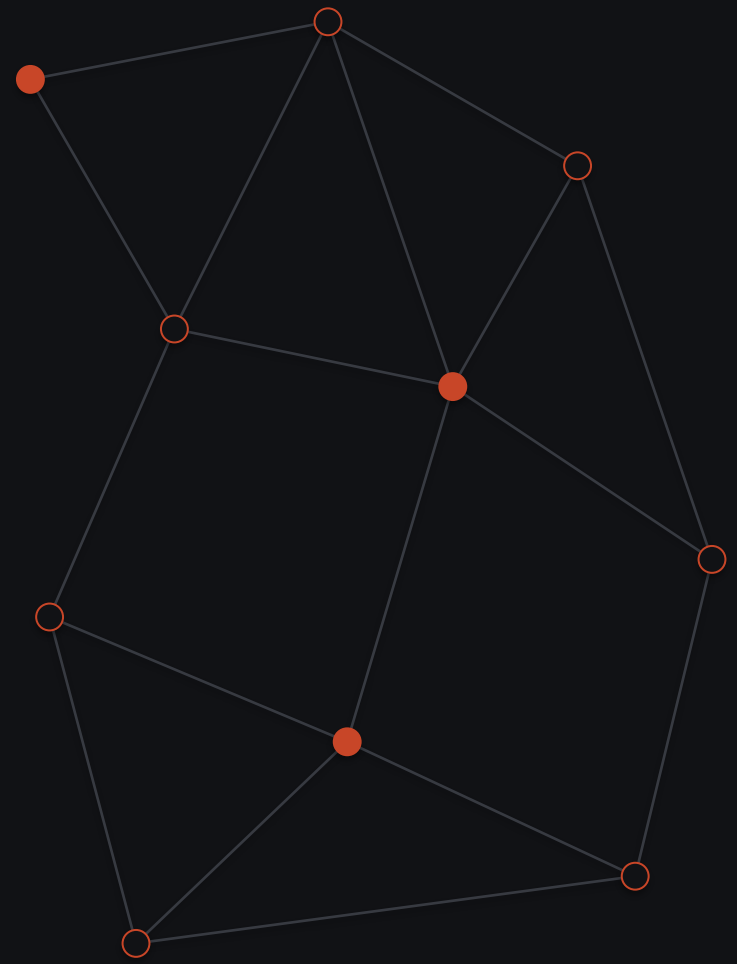
CSC03P3 PROJECTS DAY PRESENTATION

ONLINE MEMORY

Forming, Retaining & Adapting Memories from Streaming Data

ARTIFICIAL IMMUNE SYSTEM FOR INCREMENTAL LEARNING



TLALI LEMPHANE

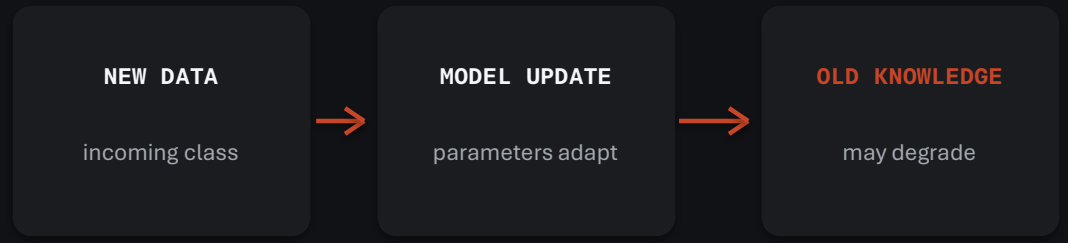
J.A.R.V.I.S.

The problem

Incremental learning asks a model to learn new classes without losing what it already knows.

AI FORGETS.

Learning a new class can degrade performance on earlier classes.



COMMON WORKAROUND: REPLAY

Keep old examples and mix them into later training.

What if previous training examples are unavailable, costly to retain, or cannot be replayed?

Research question & gap

Continual learning has established forgetting and replay as central design concerns; immune-inspired learning offers a different memory mechanism.

KNOWN PROBLEM

Sequential training can overwrite earlier knowledge.

KNOWN APPROACHES

Replay-based methods and regularization approaches address forgetting through different.

PROJECT FOCUS

Can evolved, labelled representations support incremental classification without replaying raw training examples?

RESEARCH QUESTION

Can an Artificial Immune System retain useful classification knowledge as classes arrive sequentially, without storing and replaying their original training examples?

Scope: implement and evaluate a CLONALG-inspired memory learner, study parameter sensitivity, and compare against an ANN with and without replay.

CONCEPT

The idea: store representations, not examples

The biological analogy motivates the algorithm; the implementation is a computational model, not a biological simulation.

ANTIGEN

An incoming labelled training example.

→

ANTIBODY

A candidate feature-space representation that adapts to the current class.

→

MEMORY CELL

A selected antibody retained with its class label.

COMPUTATIONAL MAPPING

Antigen	→ input example
Affinity / fitness	→ closeness to assigned target
Clonal mutation	→ candidate adaptation
Persistent memory	→ labelled representation

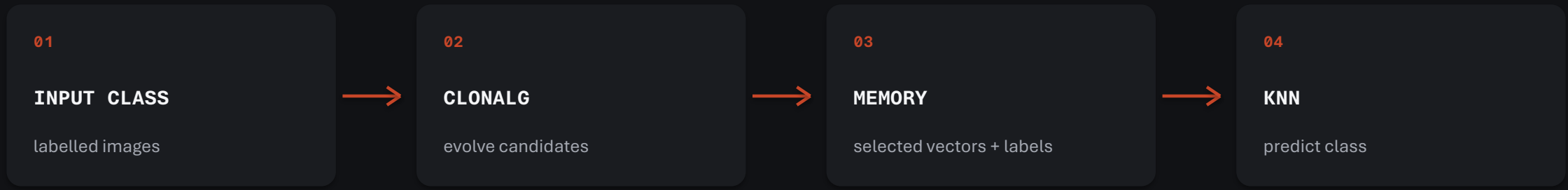
DESIGN GOAL

Retain useful representations

SYSTEM

Solution architecture

The pipeline separates class-wise learning from prediction using the retained memory.



LEARNING PATH

Training examples are used to evolve class representations.

EVALUATION PATH

Held-out test data measures performance; it does not update memory.

How memory forms

Each class is trained independently; its selected memory cells are appended after that class finishes.

01 **INITIALISE** — Seed a population from current-class examples, with noise.

02 **ASSIGN** — Each antigen is owned by its nearest antibody (Euclidean distance).

03 **ADAPT** — Build mean targets; clone selected owners towards targets with random mutation.

04 **SELECT** — Retain antibodies that own the most antigens; break ties by nearest-antigen distance.

```
Fitness during evolution = 1 / (1 + distance between an owning antibody and its target).
```

EVALUATION

Experimental design

Two complementary studies: tune the AIS, and test the role of replay in a conventional ANN.

DATA STREAM

20 classes in order: MNIST 0–9, then Fashion-MNIST 10–19.

CLONALG STUDY

19 configurations; 15 epochs per class; seeds 7 and 42.

ANN STUDY

15 epochs per class; seeds 7, 21 and 42.

No replay vs replay buffers of 100, 500, 1,000 and 2,000 examples.

MEASURES

Overall accuracy — all seen test classes

Balanced accuracy — mean of per-class accuracy

New-class recall — learning the latest class

Per-class history — retention across stages

Experiment 1: replay changes retention

Final-stage mean overall accuracy across three seeds; ANN is introduced to classes sequentially.



KEY PATTERN

~100%

latest-class accuracy

~0.01%

previous-class accuracy without replay

Without replay, the model learns the newest class but catastrophically forgets earlier ones. Larger buffers generally improve retention in this experiment.

Reported mean ± sample SD: no replay 5.01% ± 0.01%; buffer 2,000 80.09% ± 4.79%.

Experiment 2: tuning J.A.R.V.I.S.

19-configuration screening study; values shown are mean final-stage accuracy across two seeds.

BEST TESTED



79.45% • batch_size = 16

85.89%

MNIST STAGE (10 CLASSES)

config_10

79.45%

FINAL STAGE (20 CLASSES)

config_10

WHAT THIS SAYS

The gain over baseline is modest: about +0.74 percentage points at 20 classes.

Selected parameter effects at the 20-class stage

Higher noise (0.30), high mutation (0.50), selection_size = 1, and k = 7 were among the weaker tested settings. The study is targeted screening, not a full factorial search.

Interpretation: what the results support

The studies answer related questions, but they are not a perfectly controlled head-to-head comparison.

ANN WITHOUT REPLAY

Final accuracy: 5.01%.

Latest class remains near 100%, while earlier classes are almost entirely forgotten.

ANN WITH REPLAY

A buffer of 2,000 examples reaches 80.09% final accuracy on the ANN test protocol.

J.A.R.V.I.S.

Best tested configuration reaches 79.45% after 20 classes, storing evolved labelled vectors rather than replay examples.

INTERPRET WITH CARE

- Memory budgets are not equivalent: ANN replay stores raw examples; J.A.R.V.I.S. stores representations.
- CLONALG screening uses two seeds; ANN results use three seeds.

The results demonstrate the value of replay for ANN retention and the feasibility of representation-based incremental memory, not superiority.

Implementation, telemetry & performance measures

Current application settings from settings.py

CURRENT GUI CONFIG

Epochs: 15 per class
Population: 20
Memory: 10 cells per class
Selection: 3 • clones: 3 • Similarity Threshold: 7.5
Mutation: 0.20 • noise: 0.10 • alpha max: 0.5
KNN: k = 5, Euclidean, distance-weighted

TELEMETRY

Training/class/batch events
Epoch: mean nearest-memory distance, population size, clones added, cells pruned
Stage evaluation: accuracy, balanced accuracy, new-class recall, per-class scores, confusion matrix

SYSTEM STRUCTURE

PyQt6 UI + QThread training worker
DataStore loads/caches OpenML datasets
CLONALG independent of Qt
Memory, training and metrics views

NON-FUNCTIONAL NOTES

Responsiveness training runs in a worker thread

Scalability memory grows with classes × selected cells

Reproducibility random seed control belongs to experiment scripts

Robustness class memory is committed after full class training

Conclusion & next steps

FORM. CLONALG evolves candidate representations for each incoming class.

RETAIN. A small set of selected vectors is persisted with class labels.

ADAPT. KNN predicts from the memory accumulated so far.

LIMITATIONS Representation and replay memory are not equivalent budgets.

NEXT STEPS Expand repeated runs; measure memory cost and class-wise retention.

Can AI learn to remember knowledge rather than data? This project demonstrates one workable route, and the next experiments needed to validate it.